

Databases and Data Management
DNDS5020

Crime Data Management

Group B

Final Report

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1 Introduction

The issue of crime remains highly pertinent both globally and within the specific context of the Russian Federation. For example, the number of homicides remains higher than in other countries (by 2019 to 7.64 homicides per 100,000 people, compared to 3.14 per 100,000 people in Belarus and the average for Eastern Europe - 4.81). However, analysing this phenomenon is challenging, since a vast amount of information for analysis is disseminated across various sources. It is typically collected by disparate authorities, resulting in a lack of centralized data.

Currently, sentence-level datasets are available that incorporate individual case characteristics, such as the gender of the victim and the judge, as well as the presence of mitigating or aggravating circumstances (Zhuchkova & Kazun, 2023). However, these datasets often lack regional-level variables essential for comprehensive crime modeling, such as unemployment rates and GDP indicators (Andrienko & Shelley, 2005), which are typically compiled by national statistical agencies. Consequently, a substantial portion of the research process is dedicated to the collection and integration of these heterogeneous data sources.

The objective of this project is to develop a comprehensive database that enables interdisciplinary analysis of criminal activity. Such a resource would facilitate the testing of complex hypotheses, such as whether more stringent regional sentencing correlates with lower crime rates – a core tenet of deterrence theory (Becker, 1968) – while simultaneously accounting for micro-level variables derived from judicial verdicts (e.g., the gender of the perpetrator, victim, or judge). Furthermore, environmental factors, such as regional average monthly temperatures, could be utilized as proxy variables to examine causal relationships.

Beyond addressing various research gaps, the integration of such data could facilitate more effective management and workload distribution within the Russian judicial system. This is particularly relevant given the significant systemic strain, as courts remain overburdened with cases (Zhuchkova & Kazun, 2023), a condition that also extends to prisons (Ibid.).

¹ Victims of Intentional Homicide per 100 000 population by year – Regional Estimates. United Nations Office on Drugs and Crime (UNODC). <https://dataunodc.un.org/content/country-list> (05.05.2024)

2 Background

2.1 Limitations

Over the past 15 years, crime rates have declined globally, a similar trend is observed across Russian regions, reflecting policy shifts aimed at legislative reform and the modernization of judicial practices (Afanasieva et al., 2019). But regardless of reduction in crime rates in recent years, the need for improved and new data management still exists, and it could significantly contribute towards the further mitigation of crime rates in Russian regions. Also that data management with inclusion of other micro entities (police deployment, pending court cases) could be used in other world countries.

The empirical studies of criminal activities in Russia indicate that the distribution of the crimes is highly heterogeneous on the geographical, temporal as well as spatial lines, particularly in the case of homicide and other violent crimes (Pridemore, 2003). Crime rates are associated with social and economic factors (Ilgova et al., 2019; Afanasieva et al., 2019). After examining regional statistics, we can say that the places where there is a high rate of unemployment, low incomes tend to have more crime and the places regions with a higher amount of money tend to have low crime rates. For example, a study by Ilgova and colleagues (2019) conducted a cross-regional detour into demographics, crime related, and economic and cultural data through official crime data by the Prosecutor General office and socio-economic data. The findings demonstrate that there is a great deal of disparity in crime rates with quality of life and unemployment measures (Ibid.). The level of alcohol consumption within the population also significantly impacts regional crime rates, particularly in relation to violent offenses (Lysova et al., 2012).

Courts are as well the contributing factors in crime rates as if courts keep piling with the pending cases the crime rates will also tend to increase significantly (Antonov, 2019). And the deployment of police in different federal states is also a contributing factor to the lower crime rates (Pridemore & Kim, 2007).

Beyond regional indicators, case-level data are instrumental in modeling criminal activity (Zhuchkova & Kazun, 2023).

Information extracted from judicial verdicts facilitates the estimation of average sentencing lengths, enabling researchers to examine their potential impact on crime rates (Becker, 1968).

However, not all studies account for this extensive range of diverse factors, a limitation that can be partially attributed to the lack of data integration (Hauner, 2009). Furthermore, the rate of corruption and low-quality governance structures end up bending the reporting of crime process and court efficiency, and thus diminishing the analytical value of the available data even more (Antonov, 2019). The integration of multi-level data from disparate sources enables a more precise evaluation of various metrics, allows for the inclusion of a broader range of indicators, and mitigates the bias introduced by institutional inefficiencies.

This project aims to create a database that overcomes these limitations by providing a clear and standardized data structure that enables analytical insights and supports analysis of crime across Russia regions, with an overall structure that is adaptable for use in a global context.

2.2 Previous work

Before delving into the solution of our primary task, we first conduct a close examination of an existing approach, identifying its strengths and limitations to inform our own practices. Data management challenges in the realm of crime statistics are pertinent to numerous countries. For instance, in Brazil, a system was developed to enhance the transparency of crime-related data (Soares et al., 2017).

As in Russia, much data is publicly available but dispersed across fragmented formats, complicating integration. The Brazilian Open Data Portal (Portal Brasileiro de Dados Abertos; <http://dados.gov.br>) publishes diverse datasets, such as population health indicators. Moreover, authors note that various cities and states also release open data, encouraging public analytics.

However, these data are often disseminated in PDF format, which hinders rapid review, visualization, filtering, and similar operations. Additionally, the published information exhibits varying levels of granularity across regions.

The authors propose a four-layer platform architecture grounded in the "social machine" paradigm: Data Source Layer: Sources including PDF crime reports and population data. Data Processing Layer: Repositories for data extraction, filtering, and storage (e.g., MySQL with Hibernate). Service Layer: APIs for basic management (e.g., filtering by gender and age) and statistics (e.g., city rankings by murders per 100,000 residents, trends by gender/age). Application Layer: Web tools for intuitive visualization and comparisons.

Given the parallels with our challenges, we adopt the concept of integrating multi-level and multi-source data (e.g., police-reported crimes, population censuses, and social indicators). Furthermore, in selecting metrics, we draw from the article's recommendations, incorporating salient dimensions such as crimes by city, by gender and age, with support for longitudinal analysis.

2.3 Data Management Problem

Data management challenges exist at the level of these datasets. The format of the data varies. There is structured data showing statistical indicators, but also text data. The merging of the two datasets needs a data management solution since the region level information for each year needs to match with the other information that exists about each case. Scalability and reproducibility require a relational database. Complex queries are planned to be applied, in parallel with expending the structure with new data.

The purpose of the proposed data management solution is to create a database structure to support interdisciplinary research on crime, sentencing, and socio-economic conditions. Potential users are academic researchers in criminology, economics, and even data science. Policy makers and institutional stakeholders interested in evaluating judicial efficiency and regional disparities might also be interested. Centralizing and standardizing heterogeneous data sources, the project reduces the time spent on data preparation and allows users to focus only on the analysis that would otherwise be difficult to be implemented.

3 Methodology

3.1 Data Sources

The project uses as a base line two already existing datasets. These complementary datasets differ in their level of aggregation, structure, and origin, but together provide the empirical foundation for an interdisciplinary analysis of criminal activity in Russia.

3.1.1 Case-Level Dataset

The first one has 20531 observations extracted from official sentencing texts for cases under Part 1 of Article 105 ("Murder") of the Russian Criminal Code. It was published on the "Pravosudie" website with data available from 2013 to 2019. This database presents an overview of all key case-level characteristics. It was compiled and made publicly available as part of a study on gender differences in judicial decision-making (Zhuchkova & Kazun, 2023). Each record corresponds to a single defendant and includes defendant and victim characteristics (gender, parental status), judicial characteristics (judge gender), case attributes (year of offence, guilty plea), and legal factors (aggravating and mitigating circumstances, recidivism type, and sentence duration in months). The data set accounts for both legal and extralegal factors that influence the severity of the sentence. The data was extracted from sentencing texts using machine learning methods.

The dataset covers approximately 45% of all cases under Part 1 of Article 105(Ibid.). This shows that courts do not publish all sentencing texts despite formal requirements and that the dataset includes only cases where Part 1 Article 105 served as the sole charge against the defendant (Ibid.). These limitations highlight challenges related to representativeness, missing data, and selection bias.

3.1.2 Regional Dataset

The second source is a panel dataset covering 76 Russian administrative regions across the years 2012–2019. Regional crime and clearance rate data were collected from the official portal of the Prosecutor General's Office of the Russian Federation (*crimestat.ru*). Socio-economic indicators, for example income inequality (Gini index), unemployment rate, and gross regional product, were obtained from the Unified Interdepartmental Statistical Information System (UISIS). This dataset has been used in prior cross-regional analyses of crime in Russia (Ilgova, 2019 & Hauner, 2009).

The regional dataset provides the macro-level context that individual case records lack. Prior research has shown that regional unemployment, income inequality, and policing capacity are significant predictors of homicide rates (Pridemore, 2003 & Pridemore 2007), and that integrating these indicators with case-level data substantially improves the analytical precision of criminological models (Zhuchkova, 2023). The combination of both datasets in a single relational structure is the core data management contribution of this project.

3.1.3 Synthetic Data

Since the two source datasets do not cover all entities required by the full relational schema, additional data was generated synthetically to populate the institutional layer of the database. This includes police departments, police officers, prisons, investigations, evidence records, appeals, and police deployment figures. Synthetic attributes are generated using the **Faker** library with a Russian locale, combined with controlled randomization to produce categorical and numerical values.

The synthetic data is structured around two priorities. The first is geographic anchoring: police departments and prisons are linked to real region-level identifiers drawn from `spatial_hierarchy`, so that every synthetic institution belongs to an actual Russian federal subject rather than an arbitrary location. The second is relational consistency with the real case data: investigations and evidence records are linked to a random sample of real case identifiers, and appeals are restricted to cases whose status was set to "Appealed". Court identifiers for appeal records are inherited directly from the associated case. Sentence-to-prison assignments in `sentence_prison` are restricted to sentences of type "Imprisonment". Police deployment figures are generated for all real region–year combinations from the 2013–2019 period, aligning with the coverage of the regional dataset.

Temporal attributes, such as investigation periods, appeal timelines, and prison entry and release dates, are generated within constrained date ranges, with end dates occurring after start dates when present. Risk assessment scores for offenders are generated within numerical ranges calibrated to the recidivism classification recorded in the case-level dataset, so that higher degrees of recidivism correspond to higher score ranges. This approach is conceptually aligned with the logic of actuarial risk instruments such as Static-99 (Hanson, 1999), without reproducing their full methodology.

The synthetic data was structured based on key structural and relational characteristics of the criminal justice system. It prioritizes consistency and completeness of the schema over realistic institutional or behavioural modeling.

3.2 Conceptual Framework

3.2.1 System Overview

The database creates a model of the full lifecycle of a criminal case. It highlights how it moves through the Russian judicial system. This lifecycle provides the step by step approach of the schema: a crime occurs and is recorded, an investigation is opened, the case is brought before a court, a sentence is handed down, and the defendant may be incarcerated and

subsequently appeal the decision. Each stage of this process corresponds to a distinct layer of the relational model, and the entities within each layer are linked through foreign keys that reflect the real-world dependencies between them.

The following figure illustrates this structure of the created database. The **case_file** table sits at the center of the schema as the main entity, connected to the judicial layer (courts, judges, appeals), to outcomes (sentences, prison assignments), and to investigations, evidence, and case participants. The spatial hierarchy covers all institutional entities, and regional statistics provides the macro-level context that enables cross-regional analysis.

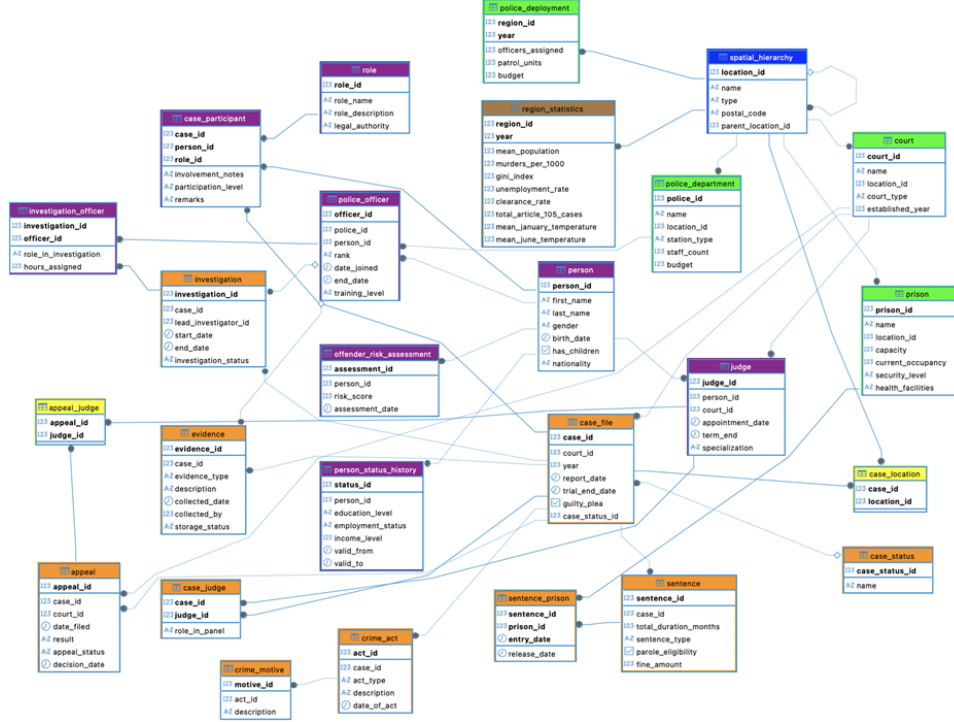


Figure 1: Relational Database

3.2.2 Entity Groups

The 26 tables in the schema are organised into seven functional layers, each corresponding to a distinct aspect of the criminal justice system.

The **geographic layer** is built around **spatial_hierarchy**, a self-referencing table that encodes the Russian administrative hierarchy from country level down to district and city. All institutional entities (courts, police departments, prisons) are related to this table, as are crime locations and regional statistics. The self-referencing design, using a **parent_location_id** foreign key, avoids the need for separate tables at each geographic level and allows the hierarchy to be extended without structural changes. A subgroup from this layer is the **regional layer**, which consists of **region_statistics** and **police_deployment**, both keyed by (**region_id**, **year**). **region_statistics** stores mean population, murders per 1,000 inhabitants, Gini index, unemployment rate, clearance rate, total Article 105 cases, and mean January and June temperatures. These indicators were chosen because prior research has consistently linked socio-economic conditions and environmental factors to regional variation in violent crime (Ilgova, 2019 & Pridemore, 2003). **police_deployment** records the number of officers assigned, patrol units, and budget per region per year, capturing the policing capacity dimension that deterrence theory identifies as a key driver of crime rates (Becker, 1968). Together these tables form the bridge between individual case records and macro-level analytical queries.

The **person layer** centres on the **person** table, which serves as the universal actor entity. A single record is created per individual and then referenced across multiple roles: as an offender, victim, lawyer, or judge via **case_participant**, as a police officer, or as a judicial appointment. The core attributes, gender, birth date, nationality, and parental status, are stored here because they are stable characteristics of an individual that do not change across cases. Socio-economic attributes, by contrast, can change over time and are therefore tracked separately in **person_status_history** using **valid_from** and **valid_to** date fields, recording education level, employment status, and income level at each point in time. This separation supports longitudinal analysis of how individual circumstances evolve in relation to criminal outcomes (Ilgova, 2019). Risk assessments are stored in **offender_risk_assessment**, which records a numerical risk score and assessment date per offender, calibrated to the recidivism classification from the source dataset.

The **law enforcement layer** captures the investigative process. **police.department** stores the station name, type, staff count, and budget, institutional attributes that allow policing capacity to be analysed alongside regional crime rates. **police.officer** records rank, training level, and employment period, attributes relevant to studying how officer experience and institutional rank relate to investigative outcomes. **investigation** stores start and end dates and investigation status, enabling the duration and resolution of investigations to be examined. The **investigation.officer** associative table records each officer's role and hours assigned, resolving the many-to-many relationship between officers and investigations and capturing individual-level contribution to each case. **evidence** captures evidence type, collection date, collecting officer, and storage status, providing a record of the evidentiary basis of each case.

The **judicial layer** connects courts, judges, and appeals. **court** stores the court name, type, and establishment year, allowing cases to be contextualised within the institutional history and jurisdiction of the court. **judge** records appointment date, term end date, and specialisation (attributes that matter because the gender, seniority, and specialisation of the presiding judge have been shown to influence sentencing decisions (Zhuchkova, 2023)). **case.judge** stores the role within the panel, distinguishing the presiding judge from associate members. **appeal** records the filing date, decision date, appeal status, and result, making it possible to track how decisions change through the appellate process and which judicial panels are associated with different outcomes.

The **case layer** is organised around **case.file** as the focal entity. It stores the report date, trial end date, year, and guilty plea (the core temporal and procedural attributes of a case). Case status is normalised into a separate **case.status** lookup table to prevent free-text inconsistencies. **crime.act** records the act type, description, and date of the act, capturing what happened and when at the level of the individual criminal event. **crime.motive** stores a description of the motive at the act level rather than the case level, reflecting the fact that different acts within the same case may be driven by different motivations (a distinction relevant to the analysis of intimate partner homicides and other relationship-driven crimes). **case.participant** records involvement notes, participation level, and remarks alongside the typed role reference, allowing the circumstances of each participant's involvement to be documented. Crime locations are recorded via **case.location**, linking cases to the geographic context in which they occurred.

The **outcome layer** records sentencing and incarceration. **sentence** stores the sentence type, total duration in months, fine amount, and parole eligibility (the attributes directly relevant to studying sentencing severity and its determinants, which is the central analytical focus of the project). **sentence.prison** records the entry date and release date for each prison assignment, with a composite primary key of (**sentence.id**, **prison.id**, **entry.date**) that allows multiple facility transfers to be recorded over time. **prison** stores capacity, current occupancy, security level, and health facility information, capturing the institutional conditions of incarceration that are relevant to modelling the burden on the Russian penitentiary system (Zhuchkova and Kazun, 2023).

3.3 Schema Design

3.3.1 Normalization

The schema is designed to third normal form (3NF), which eliminates transitive dependencies and ensures that each attribute depends only on the primary key of its table. Due to this redundancy is reduced and prevents anomalies. The trade-offs between normalization and usability is clear from the design choices. Compared to the schema proposed in the project proposal, as seen in the previous section, several new entities were created. Besides this improvement also extra modifications in the entities were applied. For example, the **case.status** table was not part of the original schema: the status was stored as a free-text **VARCHAR** column in **case.file**, which would have permitted inconsistent values. Normalizing it into a separate table and enforcing the reference through a foreign key ensures that only valid, consistent status values can be recorded. The same strategy was applied also to the **role** table, which replaces the text role descriptions in **case.participant** with a structured reference to Russian procedural law. In addition, the **success.rate** column was excluded from the **investigation** table, this metric is fully derivable from existing data. Since there already exists, the associated case resulted in a guilty plea or a completed trial, it would create a derived attribute that could become inconsistent with the underlying records over time. These are only some examples of small improvements, but a significant number of changes was made.

3.3.2 Data Types

Data types is a very important aspect in the design of a schema. In the presented structure, all primary keys use **SERIAL** and **BIGINT** is used for population counts, since these are values exceeding the **INT** range. Names and categorical labels use **VARCHAR** and the existing text fields use **TEXT**. All date fields use the **DATE** type and boolean attributes, such as guilty plea, parole eligibility, parental status, use **BOOLEAN**. Monetary values and statistical rates are stored as **NUMERIC**, which provides exact decimal arithmetic and avoids the rounding errors associated with floating-point types. Budgets and fines use **NUMERIC(12,2)** or **NUMERIC(15,2)**, percentage rates use **NUMERIC(5,2)**, the Gini index uses **NUMERIC(5,3)**, and the murders-per-1,000 rate uses **NUMERIC(6,3)**.

3.3.3 Constraints

Constraints were one of the most demanding parts in the design. These are applied throughout the schema to enforce data validity.

Primary key constraints: implemented as a single column `SERIAL` keys for independent entities and composite keys for associative tables. The composite key on `sentence_prison` (`sentence_id`, `prison_id`, `entry_date`) is very important because including `entry_date` as a key component allows multiple prison transfers to be recorded for the same sentence without any collision or other errors.

Foreign key constraints: enforce referential integrity and applies deletion actions to make stronger the dependency between entities. Another important step is the cascade deletion (`ON DELETE CASCADE`) which is used where a child record has no meaning without its parent (e.g. `crime_act` relative to `case_file`). Restrict (`ON DELETE RESTRICT`) is used where the parent should not be deleted while dependents exist (e.g. a court referenced by active cases). In the end, set null (`ON DELETE SET NULL`) is used where the dependent record remains valid even if the referenced entity is removed (e.g. an investigation that loses its lead officer).

CHECK constraints: used for the date fields recording past events which should be bounded by `CURRENT_DATE`. The constraint is applied to date ranges for a logical order or null if it is still open. The `established_year` constraint in `court` uses a dynamic bound against `CURRENT_DATE` rather than a fixed year, making it robust to future use of the schema. The Gini index is bounded between 0 and 1, and unemployment and clearance rates between 0 and 100. A cross-column constraint on `prison` prevents `current_occupancy` from exceeding `capacity`.

UNIQUE constraints: prevent duplicates records. The combination (`name`, `location_id`) on `police_department`, `court`, and `prison` ensures that no two institutions of the same name appear at the same location. The `UNIQUE` constraint on `person_id` within `police_officer` and `judge` highlights the one-to-one relationship between a person and their institutional role.

3.3.4 Relationships, Cardinality, and Ordinality

The schema follows standard relational normalisation principles, with relationships implemented through foreign keys and associative tables for many-to-many cases.

One-to-many relationships: a single `person` record can have many `person_status_history` entries over time. A court employs many judges, and a police department employs many officers. A case file generates many crime acts, each of these acts has multiple motives. A case implies many investigations that produce many pieces of evidence, and result in multiple sentences. A sentence can map to multiple prison records through `sentence_prison` as the person is transferred between facilities. The `spatial_hierarchy` table is also self-referencing in a one-to-many relationship, since each location can have one parent, but many children.

Many-to-many relationships: imply associative tables and each has a composite primary key.

- `case_participant` (`case_file`, `person`, `role`): links persons to cases with a typed role. The primary key is composite (`case_id`, `person_id`, `role_id`), meaning the same person can have multiple roles on the same case (e.g. both witness and civil plaintiff). It also uncludes `involvement_notes`, `participation_level`, and remarks as extra attributes
- `case_judge` (`case_file`, `judge`): connects judges to cases. Primary key is (`case_id`, `judge_id`). Carries `role_in_panel` to distinguish the presiding judge from associate panel members. This is separate from the judge's court assignment, because a judge belongs to a court, but may hear cases across multiple sessions.
- `case_location` (`case_file`, `spatial_hierarchy`): links cases to one or more geographic locations (e.g. the crime scene and the arrest location). Primary key is (`case_id`, `location_id`)
- `investigation_officer` (`investigation`, `police_officer`): connects officers to investigations they contributed to. Primary key is (`investigation_id`, `officer_id`). Carries `role_in_investigation` and `hours_assigned` as meaningful attributes.
- `appeal_judge` (`appeal`, `judge`): links judges to the appeals they adjudicated. Primary key is (`appeal_id`, `judge_id`)
- `sentence_prison` (`sentence`, `prison`): connects sentences to prison facilities over time. Primary key is (`sentence_id`, `prison_id`, `entry_date`). The important element included `entry_date`. It allows multiple transfers between facilities to be recorded for the same sentence. It also carries `release_date`, which is nullable for currently incarcerated individuals

One-to-one relationships: appear where a person can hold only one instance of a given institutional role. The `UNIQUE` constraint on `person_id` within `police_officer` and `judge` makes sure that each person can hold maximum one officer record and one judge record at the same time, and it still allows those records to carry distinct institutional attributes.

Ordinality: depicts if participation in a relationship is mandatory or optional. It is implemented through nullability, so mandatory foreign keys are declared `NOT NULL`. For example, a case file must reference a court, and an officer must belong to a department. References can be left nullable. One such case can be that the lead investigator on an investigation may

be unknown or removed, so `lead_investigator_id` is nullable with an `ON DELETE SET NULL` action. Same approach appears also in the case of `term_end` on `judge` and `end_date` on `police_officer` which are nullable in order to represent currently active individuals.

3.3.5 Indexes

New indexes were added on the most frequently joined and filtered columns, in addition to the primary and foreign key indexes that PostgreSQL creates automatically. These cover the foreign keys in `case_file` (`court_id`, `case_status_id`), the person (`person_id`) reference in `case_participant`, and the case (`case_id`) references in `crime_act`, `investigation`, and `evidence`. Indexes were also added on `police_officer.police_id` and `judge.court_id`, since both appear in queries that aggregate officer and judge statistics by institution. These indexes are very relevant given that the planned analytical queries involve many `JOIN` statements between table, across all of these dimensions, the presented seven layers, at the same time.

Table 1: Summary of the entities

Entity	Layer	Role in schema
<code>spatial_hierarchy</code>	Geographic	Self-referencing location tree (country → district)
<code>region_statistics</code>	Geographic	Annual socio-economic indicators per region
<code>police_deployment</code>	Geographic	Annual officer deployment figures per region
<code>person</code>	Person	Role → offender, victim, judge, officer
<code>person_status_history</code>	Person	Longitudinal socio-economic attributes
<code>offender_risk_assessment</code>	Person	Risk scores calibrated to recidivism type
<code>court</code>	Judicial	Judicial institutions linked to locations
<code>judge</code>	Judicial	Person appointed to a court with term dates
<code>case_judge</code>	Judicial	M:N → judges assigned to specific cases
<code>appeal</code>	Judicial	Challenges filed against case decisions
<code>appeal_judge</code>	Judicial	M:N → judges assigned to appeal panels
<code>case_file</code>	Case	Main entity (central point of the schema)
<code>case_status</code>	Case	Lookup table for normalised case statuses
<code>case_participant</code>	Case	M:N → persons linked to cases via typed roles
<code>role</code>	Case	Lookup table grounded in Russian CPC articles
<code>case_location</code>	Case	M:N → geographic locations linked to cases
<code>crime_act</code>	Case	Individual criminal acts within a case
<code>crime_motive</code>	Case	Motives recorded at the act level
<code>police_department</code>	Law	Institutional units linked to locations
<code>police_officer</code>	Law	Officers employed by departments
<code>investigation</code>	Law	Investigations linked to cases
<code>investigation_officer</code>	Law	M:N → officers assigned to investigations
<code>evidence</code>	Law	Evidence items collected per case
<code>sentence</code>	Outcome	Judicial outcome of each case
<code>sentence_prison</code>	Outcome	M:N → prison assignments over time
<code>prison</code>	Outcome	Correctional facilities linked to locations

3.4 Implementation

3.4.1 Repository: project work

The database is implemented in PostgreSQL, chosen for its robust support for relational modelling, complex constraints, and the analytical query patterns planned for this project. The schema is organised under a dedicated `project` schema namespace. All table creation, constraint, and index creation are implemented as SQL scripts that can be run in sequence to reproduce the full schema from scratch. All the documents related to this project can be found in the GitHub repository.

GitHub: Crime Data Management Repository

This repository contains two files needed to reproduce the database from scratch:

The first is `crime_database.sql`, which creates the full schema. To run it, open DBeaver, connect to your PostgreSQL

instance, and execute the file: SQL Editor → Open SQL Script. This creates all 26 tables under the `project` schema, including constraints and indexes, in the correct dependency order.

The second is `populate.db.final.ipynb`, a Jupyter notebook that loads the two source datasets, maps their variables to the schema, generates synthetic records for the institutional tables, and inserts all 316,073 rows into the database. Before running it, update the connection block at the top of the notebook with your host, database name, username, and password, and ensure that `105.1.regression.data.en.csv` and `Registru.xlsx` are in the same directory. The required Python packages are `pandas`, `openpyxl`, `faker`, `sqlalchemy`, and `psycopg2-binary`, all installable via `pip`.

Recommendation: run the cells in the provided order, since the insertion sequence follows the foreign key chain, and these steps are very important, no step can be skipped.

Once populated, the database can be queried, for example for this project DBeaver was used. The README in the repository includes four example queries in the `queries.sql` file.

3.4.2 Demonstration of work

Query 1: Average time from crime act to verdict by crime type. The first query demonstrates how the schema can be used to monitor the efficiency of the judicial process at different stages.

Query purpose: Monitoring the duration of the judicial process from the date of the crime (or act registration) to the issuance of the final verdict. This helps identify “bottlenecks” in the justice system:

- Acceleration of case processing for mass articles (e.g., Article 228 of the Criminal Code of the Russian Federation — drug crimes) may indicate a formal approach without proper evidence analysis.
- Delays in rare or complex cases signal the need for optimization (additional judges, simplified procedures).

Query output: a table grouped by act types (Criminal Code/Administrative Code articles):

- average time in days and months from `date_of_act` to `trial_end_date`;
- total number of completed cases per type;
- sorted by descending processing time (longest first).

Filters are used in the request: we consider only completed cases (`trial_end_date` IS NOT NULL), as well as those cases where there are no violations of logic in the deadlines (valid dates (`date_of_act` IS NOT NULL, the verdict was issued after the crime was committed)).

Since our database is currently populated only with homicide data, the output contains only one row. In the future, the database will scale to include additional case types.

Table 2: Query 1 results: Average processing time by crime type

act_type	avg_days_to_verdict	avg_months_to_verdict	total_cases
Homicide	345.3	11.3	19512

Query 2: Different statistics about cases by region (for example income, sentence length).

Query purpose: Demonstrate work with spatial hierarchy (`spatial_hierarchy`) and aggregation of statistics at different levels (regions, cities, districts).

Such queries are useful for:

- analyzing various correlations and causal relationships, e.g., between socio-economic indicators and punishment severity or temperature impact on crime rates in regions.
- detecting regional disparities in law enforcement, e.g., whether richer regions issue softer sentences on average.

This query example outputs statistics at region level (filter: `spatial_hierarchy.type = 'region'`). In this case, we are showing only the first lines of the output.

Query output: table by Russian Federation regions:

- average income of convicts (`avg_income` from `person.status.history`);
- average sentence length in months (`avg_sentence_length`);
- sorted by descending average income (regions with most affluent convicts at top).

Table 3: Query 2 results: Statistics about cases by region

region_name	avg_income	avg_sentence_length
Republic of Dagestan	57 570.658	103.41
Smolensk region	56974.042	98.516
Republic of Mordovia	56094.978	107.875

Query 3: Police effectiveness and clearance rates by region/year. Query purpose: Demonstrates how to use the schema to track police effectiveness — Does increasing the number of police officers (officers_assigned) lead to higher clearance rates (clearance_rate)? Specifically, tests deterrence theory — Does high probability of arrest reduce crime levels in the long term?

Query output: panel table (region $\times$ year):

- year (year);
- region ID (region_id);
- case clearance percentage (clearance_rate);
- number of police officers per region (officers_assigned);
- sorting (first by years, then by regions).

These data are ready for export to Python/R for fixed effects regression by region to test deterrence hypothesis. The example shows only the first output rows.

Table 4: Query 3 results: Police effectiveness and clearance rates by region/year

year	region_id	clearance_rate	officers_assigned
2013	2	7.73	1528
2013	3	11.18	171
2013	4	6.67	2442
2013	5	15.87	889

Query 4: Average judge caseload by court. Query purpose: This query directly addresses one of the key social problems of the Russian judicial system — chronic court overload.

Such query is necessary for:

- resource planning (where additional judges/assistants are needed);
- process optimization (redistributing cases from overloaded courts);
- political decisions (justifying court budgets);
- forecasting (load trends for predictive analytics of peak loads).

Query output: table by courts:

- court name (court_name);
- average number of open cases per judge (avg_ongoing_load);
- sorted by descending load (most overloaded courts at top).

This query used a subquery that calculates the workload of individual judges by courts. This is an intermediate table (CTE-like) that is then aggregated by courts.

Table is filtered for active cases only (trial_end_date IS NULL, status 'Open'). We also use LEFT JOIN to account for judges without cases at the moment. Shows only first query rows.

3.4.3 Data Population Pipeline

The database population is implemented in the jupyter notebook previously presented, as a Python pipeline (`populate_db_final.ipynb`), using **pandas** for data loading and transformation, **Faker** for synthetic data generation, and **SQLAlchemy** with the native PostgreSQL COPY protocol for high-speed insertion. Synthetic data generation uses the **Faker** library with a Russian locale

Table 5: Query 4 results: Average judge caseload by court

court_name	avg_ongoing_load
Oktyabrsky District Court of Ulan-Ude	6
Central District Court of Komsomolsk-on-Amur	5
Belovsky City Court	5

(**ru_RU**) for person names and a US locale (**en_US**) for other generic fields such as sentences and dates. A fixed random seed (**seed = 42**) is applied to both **random** and **numpy.random** at the start of the pipeline to ensure that synthetic data is fully reproducible.

The total volume of data inserted across all 26 tables is approximately 316,073 rows. The pipeline maps the source datasets to the schema as follows. From the case-level CSV, each of the 20,531 rows produces one case record, three person records (offender, victim, judge), one crime act, one motive, one sentence, two case participant records, and one case judge record. Court records are derived from the 1,924 unique court names in the CSV. From the regional Excel file, 608 region-year combinations are inserted into **region_statistics**. Synthetic records (100 rows per institutional table) are linked to real region IDs, case IDs, and court IDs to maintain referential consistency.

Table 6: Summary database population by table

Table	Rows	Source	Inserted data
spatial_hierarchy	2,098	Mixed	Real regions + court names from CSV
region_statistics	608	Real	Excel (76 regions $\times$ 8 years)
police_deployment	539	Mixed	Real region-year combinations
person	61,693	Mixed	Gender from CSV; names synthetic
person_status_history	61,593	Mixed	Period anchored to case year
offender_risk_assessment	20,531	Mixed	Recidivism type from CSV; score synthetic
court	1,924	Real	Unique court names from CSV
judge	1,924	Mixed	Gender from CSV; dates synthetic
case_file	20,531	Real	CSV (Zhuchkova & Kazun, 2023)
case_status	5	Synthetic	Controlled vocabulary
role	10	Synthetic	Russian CPC articles
case_participant	41,062	Real	Offender + victim per case
case_location	20,531	Real	One location record per case
case_judge	20,531	Real	One judge per case
crime_act	20,531	Real	Relationship type from CSV
crime_motive	20,531	Real	Mitigating circumstances from CSV
sentence	20,531	Real	Sentence duration from CSV
police_department	100	Synthetic	Faker + real region IDs
police_officer	100	Synthetic	Faker (Russian locale)
prison	100	Synthetic	Faker + real region IDs
investigation	100	Synthetic	Sampled from real case IDs
investigation_officer	100	Synthetic	Linked to synthetic officers
evidence	100	Synthetic	Sampled from real case IDs
appeal	100	Synthetic	Sampled from Appealed cases
appeal_judge	100	Synthetic	Linked to real judge records
sentence_prison	100	Synthetic	Linked to real sentence records
Total		316,073 rows	

3.4.4 Schema Evolution

The schema presented here represents the fourth iteration of the design. The initial ER diagram proposed in the project proposal contained five core entities (cases, persons, roles, regions, and courts) directly reflecting the structure of the two source datasets. Until the final schema was achieved, the ER diagram went through multiple iterations. From the proposed schema, a too extended structure was developed, exceeding the scope of the intended analysis. After these problems were identified, a schema with less entities was created. The final schema was achieved after the full normalization was applied.

The final schema extends the original baseline to a total of 26 tables, expanding coverage to represent the full criminal justice lifecycle.

Some of the additions and the most significant ones relative to the proposal are the normalization of `case.status` into a lookup table, the introduction of `case.judge` to formalize the judge–case link (many-to-many relationship essential for the gender bias analysis described by Zhuchkova and Kazun (2023)), the addition of `crime.act` and `crime.motive` to provide act-level granularity, the introduction of `sentence.prison` to support transfer tracking, the addition of `offender.risk.assessment` to record structured risk scores, and the expansion of the geographic layer to include `police.deployment` as a standalone table rather than an attribute of the region. Each of these additions addresses either a normalisation requirement, an analytical need, or a data integrity concern identified during implementation.

4 Discussion

The crime data management system created within the framework of this project deals with one of the long-standing issues in criminology-related studies: the lack of integration of judicial, institutional, and regional data among incompatible sources. The database allows interdisciplinary analysis of the case-level sentencing data with the regional socio-economic indicators in a single relational schema, which could not be done in the past with gigantic preparation. The schema is not merely a storage tool, but a research instrument whose design encodes the analytical questions it is intended to address. The modeling decisions, such as the normalization of the person records into one entity that is shared across roles, to the division of the spatial hierarchy into a self-referencing table are directly connected to the kind of multi-level, multi-actor query required by criminological research. The closest analytical tool is deterrence theory testing. According to Becker (1968), perceived severity and certain punishment were hypothesized to be a reaction to criminal behaviour. This is directly modeled in the database: one can query sentencing length, recidivism type, and regional clearance rate at the region-year level, or whether regions with a high historical sentencing length or clearance rate have lower subsequent murder rates per thousand population. The data (2013–2019) are longitudinal, which allows for difference-in-differences analysis of policy changes implemented during this period, enabling causal interpretation where such changes can be treated as natural experiments. The institutional capacity and the effect on judicial results are also discussed in the context of the schema, and Antonov (2019) and Pridemore and Kim (2007) pose this question in relation to the institutional changes in Russia. The workload of courts can be measured by the number of cases pending in a court, the duration of sentencing, and the rate of success of appeal can be used to test the hypothesis that there will be more standardized and heuristic-based sentencing in congested courts.

The institutional analysis is furthered to more specific 2013-2019 window with the use of police deployment data which are joined with regional clearance rates and variables previously unavailable to researchers. The correlations between the structural deprivation and crime reported by Ilgova et al. (2019) can also be evaluated based on the socio-economic factors in the region statistics: Gini index, unemployment rate, mean temperature, on a more recent and comprehensive data set. These capabilities are illustrated in four analytical queries that were used throughout the project presented in section 3.4.2 .

5 Limitations

The database contains a number of significant limitations that limit the extent of conclusions that can be made out of analyses made on it. Some of these limitations exist in the source data, and the rest is associated with the constraints of the existing implementation. It calls on them to be aware in order to be responsible in using the system.

5.1 Representativeness and Selection Bias

The partial coverage of the main case-level data is the most crucial constraint. The data encompasses about 45 percent of all the murder cases related to the Articles 105 Part 1 of the Russian Criminal Code during the years 2013-2019 (Zhuchkova and Kazun, 2023). Sentencing texts are legally mandatory that courts be published but often fail to do so, forming a non-random distribution of missingness. It also eliminates any instances that involved Article 105 as one of a series of charges against the accused, which eliminates a significant number of the most complicated and serious cases those that involve other counts of assault, robbery, or organised crime. The resulting sample probably will over-represent single-perpetrator, single-victim homicides and under-represent those that are part of larger criminal behavior. Such a mechanism of selection brings systematic bias to any study of the severity of sentencing or judicial behaviour. When utilizing this database, researchers should take into consideration the likelihood that the observed patterns such as a mean sentence length, a ratio of guilty pleas, a rate of mitigating circumstances may have been caused by the characteristics of the observed sample as opposed to the actual population of homicide cases. One way of evaluating the strength of findings would be sensitivity analyses that limit the sample to areas with greater levels of publication compliance.

5.2 Institutional Reporting Bias

A number of variables in region statistics, including clearance rate and total number of article 105 cases are self-reported by the institutions in which they measure performance. Antonov (2019) reports the institutional motivation of Russian law enforcement and prosecutorial agencies to overstate the clearance rates and understate the number of registered crimes. This reporting bias was found by Hauner et al. (2009) to be a confounding variable in the cross-regional studies of the Russian data on crime. The clearance rates and official crime figures in this database should be regarded as proxies, which are subject to institutional distortion, and analyses that depend on them as outcomes variables should have robustness checks with other measures where feasible.

5.3 Russia-Specific Schema Design

Though the project mentions a desire to build a schema which can be adapted to other nations, the existing information material is highly Russia-centric. The classification of the article (Article 105 Part 1 of the Russian Criminal Code), the system of the regional administrative codifying (the 89 federal subjects), the typology of recidivism types that are established in the Russian legislation, and the social-economic indicators that are based on the Russian statistical agencies all indicate the institutional context of the legal system of a particular country. To enable comparative analysis across countries, an additional mapping layer (a shared taxonomy of types of offences, roles, administrative units, etc) is needed which is not presently represented in the schema. This is not a trivial harmonisation project and should be approached as a separate future project and not just an extension.

6 Future work

From an analytical perspective, the database could support richer research questions by examining crime motives alongside environmental factors such as temperature or unemployment, analysing interactions between multiple actors within the justice system, and testing whether investment in police training or regional infrastructure correlates with lower crime rates. The schema is also designed to be adaptable beyond Russia, enabling future comparisons across countries or crime types.

From a technical perspective, the most immediate extension would be moving towards a fully operational system with real-time data ingestion and role-based access control. The temporal tracking already implemented for persons could be extended to other entities such as prison capacity and police resources, supporting more advanced longitudinal analysis. A hybrid architecture combining PostgreSQL with NoSQL systems (Elasticsearch for full-text search over evidence and investigation notes, and Neo4j for relationship and network analysis between offenders, victims, and cases) would improve both scalability and analytical depth. Finally, machine learning models could be integrated to predict recidivism risk, detect patterns in crime data, and support decision-making in investigations

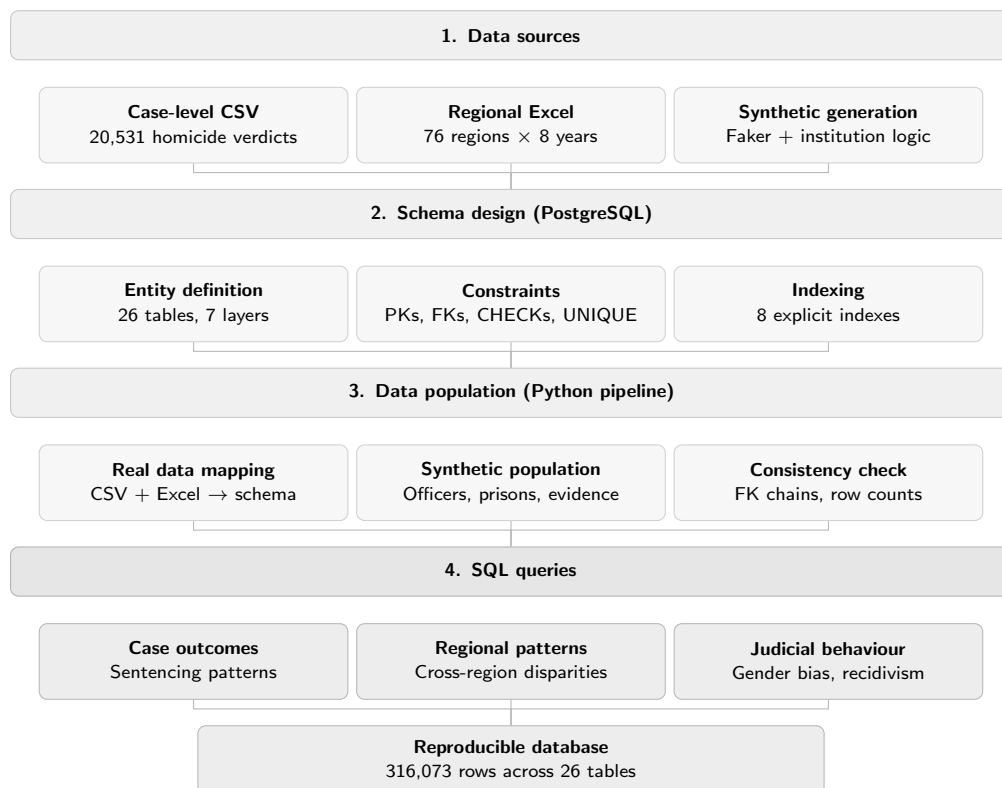
7 Conclusion

The project was intended to solve an actual data management problem: the disaggregation of crime, judicial, and regional data which could not be asked at once across a variety of sources before. The ultimate system achieves that aim and even goes beyond the extent of the initial proposal. The schema commenced with five central entities and through three reported extensions to 26 tables fully normalised with primary and foreign key constraints on all tables, CHECK constraints on key fields of importance and parent before child sequential insertion constraints on all 26 tables. The two actual data sets, 20,531 judicial cases and 608 region-year observations were successfully joined with region and year identifiers as the common join key. The extended schema consists of formalizing eight kinds of institutional actors described in the original data sets, but never structured: courts, judges, police departments, police officers, prisons, investigators, spatial hierarchy, and offender risk assessments. There are 316,073 records in the database.

The greatest learning of the team was that the decisions that are important in a data management project are made even before a SQL is written. The primary key strategy, the order of insertion and the scope of the schema were much more consequential than expected. Revisions of the schema, following loading of the initial data, caused cascading errors in dozens of dependent tables - a direct result of failing to deliver fully on the promise made by the proposal to complete the ER diagram prior to implementation. The price of that debugging time was paid by the team. The lesson about the active demonstration of data quality, but not its assumed existence, was also critical: the verification queries developed at the project conclusion are not a post-hoc audit, but the contribution of the system and should be re-executed after each subsequent data load. A database that has not been checked does not give any indication of its correctness itself, no matter how well it was constructed. The only thing left is to put the database into practice against the substantive questions that inspired the creation of the database. The facilities created here are not the end of the research. It is the place where the research can at last commence, and the worth of this project is that it has been able to do so: data on crime which previously could only be found in fragments now has a single, unified, interrogative voice.

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Appendix: End-to-end project workflow. Stage 1 covers the two real datasets and the synthetic generation layer. Stage 2 implements the relational schema in PostgreSQL. Stage 3 populates the database using a Python pipeline. Stage 4 runs analytical SQL queries on the populated database.